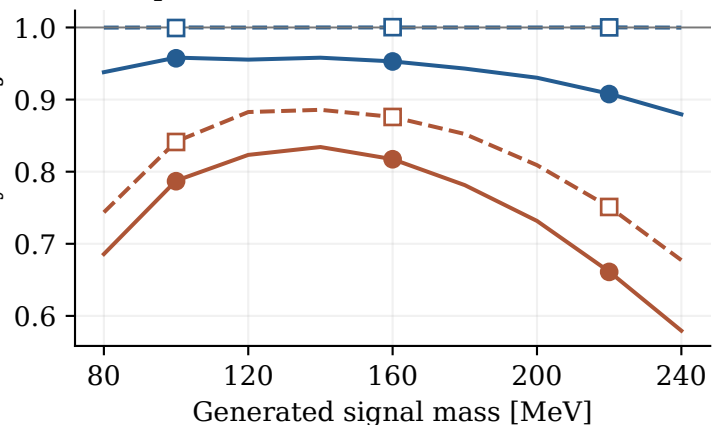


— Direct MC, starter window; contaminated — Gaussian, historical window; contaminated
- - Direct MC, starter window; clean training - - Gaussian, historical window; clean training

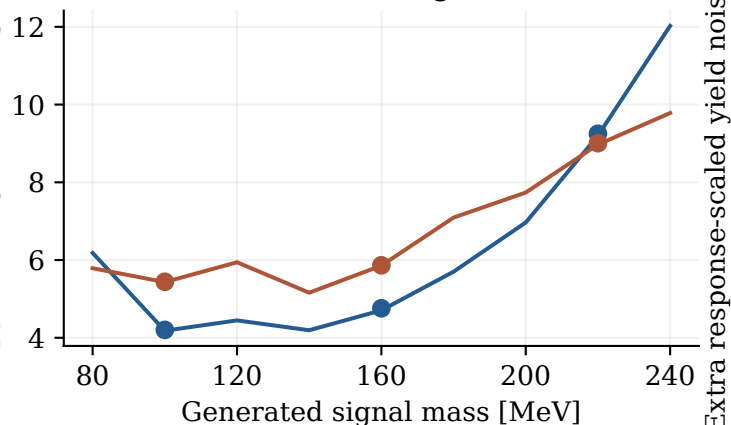
Mean added fitted yield / injected yield A

A. Response with and without contamination



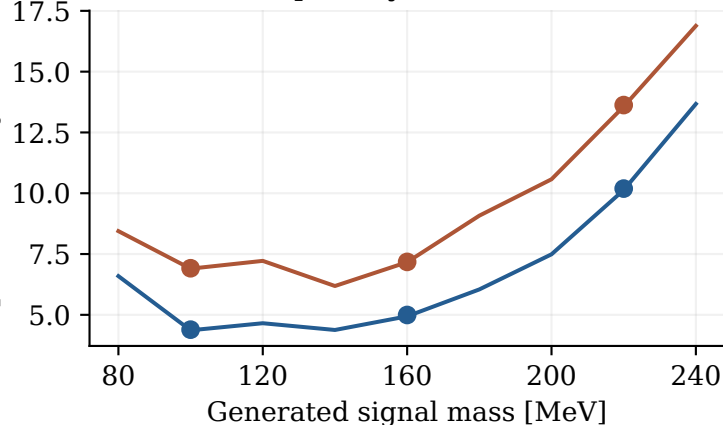
Yield suppressed by GP training [% of A]

B. Isolate the effect of signal in sidebands



Extra response-scaled yield noise [%]

C. Precision penalty from contamination



How to read: clean training removes only the injected signal from GP training bins; the signal-fit counts, template and window are identical.

Lines: deterministic mean-count spectra at nine masses, not toy means. Markers: 100 paired saved evaluation toys at 100, 160 and 220 MeV.

A/B bars: sample standard errors of paired means. C bars: 95% intervals from 2,000 whole-toy bootstrap resamples; masses/methods stay paired.

Panel B = $\text{mean}(\text{Ahat}_{\text{clean}} - \text{Ahat}_{\text{contaminated}}) / A$. Panel C = $100 \times (\text{R}_{\text{clean}} / \text{R}_{\text{contaminated}} - 1)$; background-only noise is unchanged.

s0 is the archived Gaussian yield-error scale. This truth-assisted control diagnoses absorption; it is not a usable data fit or a measured upper-limit gain.